

MOHAMED FAWZI YOUSEF MOHAMED

Senior Agentic AI Engineer & Solutions Architect

Cairo, Egypt | +249 91 559 2263 (WhatsApp/Roaming) |
dr.mohamed.fawzi.y@gmail.com

[GitHub](#)  | [Live Portfolio](#) 

PROFESSIONAL SUMMARY A Medical Doctor (MBBS) transitioned into a Senior Agentic AI Engineer and Solutions Architect. Specializes in architecting stateful multi-agent workflows, LLM orchestration, and Arabic NLP (Modern Standard & Regional Dialects). Proven track record of independently deploying low-latency Voice AI Agents and secure, scalable HealthTech systems.

TECHNICAL SKILLS

- **AI & Agentic Architectures:** Multi-Agent Workflows, LLM Orchestration, Stateful Agents, Prompt Caching, Voice AI (Vapi, ElevenLabs), Local LLMs (Ollama).
- **Backend & Infra:** Python, FastAPI, Node.js, Supabase, Vercel, RESTful APIs, Vector Databases.
- **Frontend:** React, Next.js, TypeScript, Tailwind CSS.
- **Languages:** Arabic (Native - MSA & Dialects, crucial for NLP localization), English (Fluent).

ENGINEERING PROJECTS & EXPERIENCE

Itminan Health (December 2025) | Product Designer & Full-Stack Developer

- Engineered a secure full-stack clinical platform using Supabase (PostgreSQL), strictly enforcing Row Level Security (RLS) and leveraging managed OAuth 2.0 / JWT session protocols.
- Developed an interactive clinical dashboard with click-responsive visualizations for real-time tracking of dynamic symptoms.
- Automated clinical workflows via structured PDF reports using TestSprite AI to maintain high code reliability and automated testing coverage.

H.I.M (Agentic Voice AI Conversational System) | Lead AI Engineer, Feb 2026

- Architected a low-latency, stateful Voice AI agent tailored for complex Sudanese dialect recognition, achieving <500ms response latency.
- Orchestrated dynamic multi-agent workflows integrating Vapi and ElevenLabs to execute autonomous conversational logic and medical triaging.
- Resolved critical challenges in real-time audio processing, ensuring continuous memory state management and high session reliability.

AI-Driven EdTech Assessment Tool | AI & Full-Stack Developer | Completed March 2026

- Prototyped and deployed an educational assessment platform for generating accurate, context-aware medical questionnaires.
- Implemented Prompt Caching and dynamic model switching, reducing API latency by ~40% and optimizing token costs.
- Executed successful User Acceptance Testing (UAT), receiving high praise for precision and speed from initial users.

HACKATHONS & PROJECTS

Mistral AI Hackathon (Feb 28 - Mar 1, 2026) - "ER Zero Latency" | Solo AI Solutions Architect

- Engineered a zero-latency Multimodal AI co-pilot for Emergency Room triage using passive CCTV surveillance.
- Programmed a CV pipeline (MediaPipe, OpenCV) to detect clinical distress signals and correlate visual data with EHRs for ACS prediction.
- Designed an event-driven 'Low-Power Standby' logic (Eco-Mode), reducing GPU consumption by 85%.
- Achieved a sub-100ms response time from visual trigger to final diagnostic alert.

EDUCATION

Bachelor of Medicine, Bachelor of Surgery (MBBS) - May 2025 Ibn Sina University (ISU), Khartoum, Sudan

- Completed degree with a concurrent self-directed focus on software engineering and AI architecture.